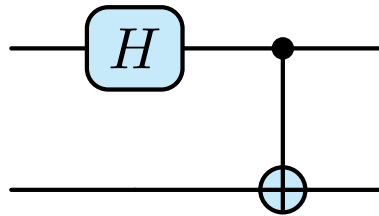


Quantum Information Processing

LECTURE NOTES

Instructor: Prof. Chiranjib Mitra



Sagnik Seth

**Dept of Physical Sciences
IISER Kolkata**

Contents

Lecture 0: Introduction

3

Lecture 0: Introduction

The course pertains to the topics in quantum information and how to process this information. Quantum computing has recently garnered a huge hype in both academia and industries and the potential applications are being studied. We will deal with the basic introduction to some aspects of quantum information theory in this course.

Although plethora of literature is available for the subject, the references for the course will be:

- **Quantum Computation and Quantum Information**; *Michael Nielsen, Isaac Chuang*. Termed as the ‘bible’ of quantum information.
- **Preskill Notes**; *John Preskill*. The link to the notes page is www.preskill.caltech.edu/ph229/
- **The Physics of Quantum Information: Quantum Cryptography, Quantum Teleportation, Quantum Computation**; *Bouwmeester, Ekert, Zeilinger*

The grading for the course consists of 20% from the midsem, 20% from two best out of three class tests, 30% from the endsem, 30% from class participation and rest 20% is contributed from either assignments and class notes.